BTDSE311N

III Semester Examination, January - 2023

B.Tech./B.Tech.+M.B.A./B.Tech.+M.Tech.(CSE,CSE-CC,CSE-CF,CSE-BDA,IT,CSE-AIML,CSE-BDAI,CSE-CMCI,CSE-ICS,CSE-ES,CSE-DSI,CSE-FSDI)

Information Theory and Coding

Choice Based Credit System (CBCS)

Time: 3 Hrs. **Maximum Marks: 60**

Minimum Pass Marks: 24

Note: All questions carry equal marks, out of which part'A' and 'B' carry 3 marks and part 'C' carries 6 marks. From each question, part 'A' and 'B' are compulsory and part 'C' has internal choice. Draw neat diagram, wherever necessary. Assume suitable data wherever necessary.

Q.1(A) Define information content of a symbol. 03

(B) Define rate of information of discrete memoryless source. 03

(C) What is source entropy? Show that entropy is maximum when all symbols are equiprobable? 06

OR

A DMS X produces one of four possible symbols during each interval having probabilities $P(x_1) = ½$, $P(x_2) = ¼$, $P(x_3) = 1/8$ and $P(x_4) = 1/8$. Find entropy of this source and information rate if source emits 250 symbols per second.

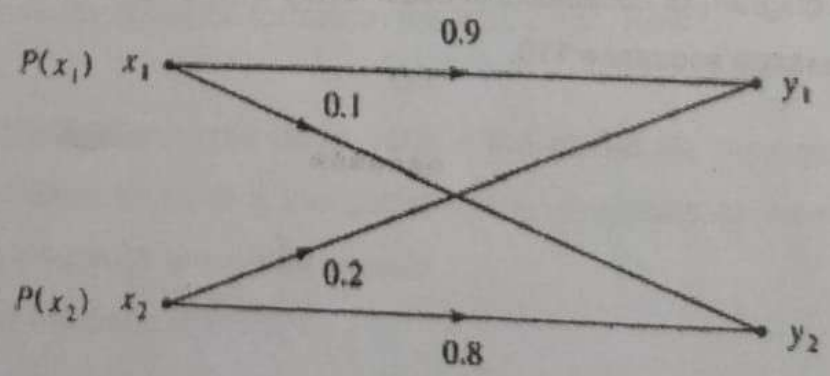
Q.2(A) What is Discrete Memoryless Channel (DMC)? Explain it. 03

(B) Define Mutual Information and Channel Capacity for Discrete Memoryless Channel. 03

(C) Consider a Discrete Memoryless Source (DMS) X that produces five symbols x_1, x_2, x_3, x_4 and x_5 with respective probabilities 0.15, 0.2, 0.15, 0.25,0.25. Construct Huffman code for X and calculate code efficiency. 06

OR

Consider a binary channel shown in the below figure.



I. Find P(y1) and P(y2) when $P(x_1) = P(x_2) = 0.5$

II. Find joint probabilities $P(x_1, y_2)$ and $P(x_2,y_1)$ when $P(x_1) = P(x_2) = 0.5$

Contd..

Q.3(A) Define channel capacity of AWGN channel. Explain how can it be increased?

(B) What is channel coding? Explain objective of channel coding.

(C) Consider a (6,3) linear block code with the generator matrix G given by

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

Suppose that the received code word is 111010. Decode this codeword by finding the location of single bit error and the transmitted message bits. 06

OR

What is continuous channel? Explain differential entropy and average mutual information in continuous channel.

Q.4(A) What is meant by Cyclic Code? Briefly compare it with convolutional codes. 03

(B) Show that code C = {0000, 0101, 1010, 1111} is a linear cyclic code. 03

(C) Consider a (7,4) cyclic code with generator polynomial $g(x) = 1 + x + x^3$.

I. Let data word **d = (1010)**, find the corresponding codeword.

II. Let the code word **c= (1100101)**, find the corresponding data word. 06

OR

Explain following codes:

I. BCH Codes

II. Goley Codes

Q.5(A) Explain source coding theorem. 03

(B) Find Transform Domain representation of Convolutional Code. 03

(C) Find generator matrix of convolutional encoder. Using generator matrix find the output code word for input message sequence 101. 06

OR

Draw state diagram of convolutional code. Using state diagram find output codeword for input message sequence 110.

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